

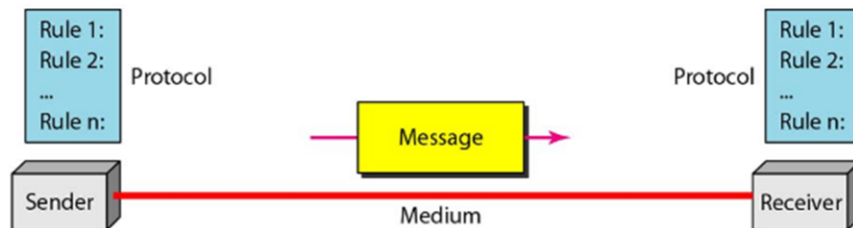
Data Communication

- What is Data Communication?

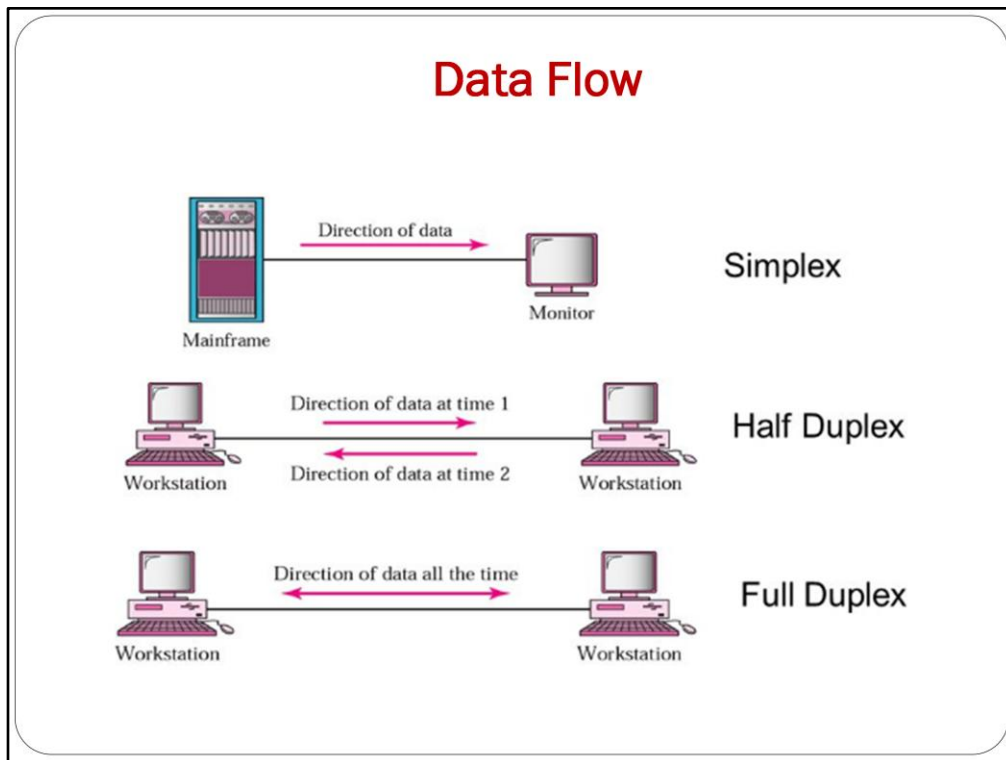
- When we communicate, we are sharing information.
- This sharing can be local or remote.
- Local communication usually occurs face to face and remote communication takes place over distance.

Data Communication

- Components of Data Communication



1. Message - The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. Sender - The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. Receiver - The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. Transmission medium - The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves
5. Protocol - A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.



Simplex:

In simplex mode, the communication is unidirectional. Only one of the two devices on a link can transmit; the other can only receive .

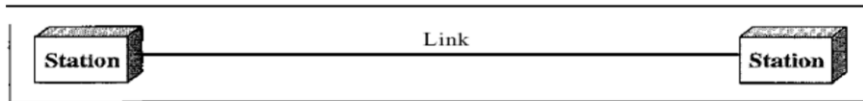
Half-Duplex:

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa.

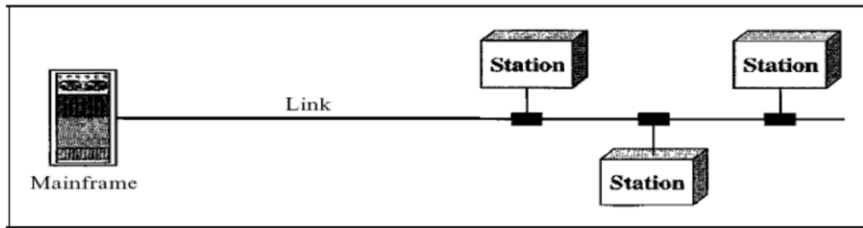
Full-Duplex:

In full-duplex both stations can transmit and receive simultaneously.

Type of Connections



a. Point-to-point



b. Multipoint

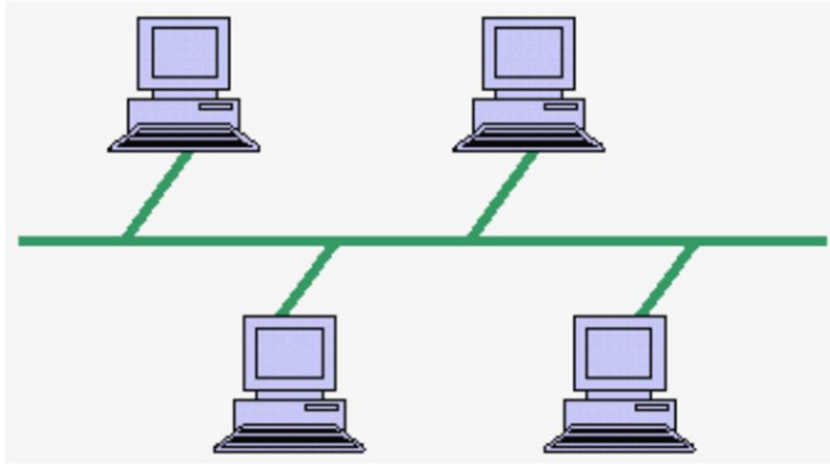
A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices.

A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a *spatially shared connection*. If users must take turns, it is a *timeshared connection*.

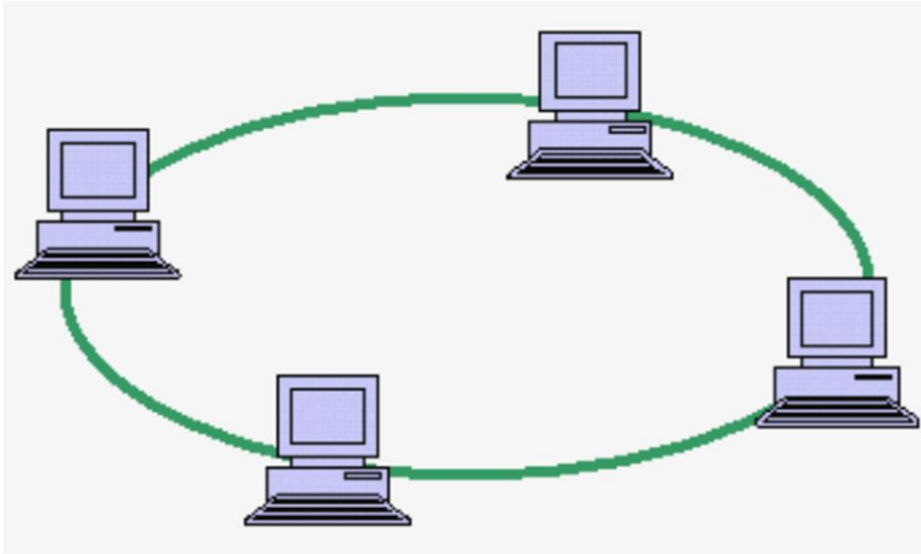
Physical Topologies

- Bus
- Ring
- Star
- Mesh

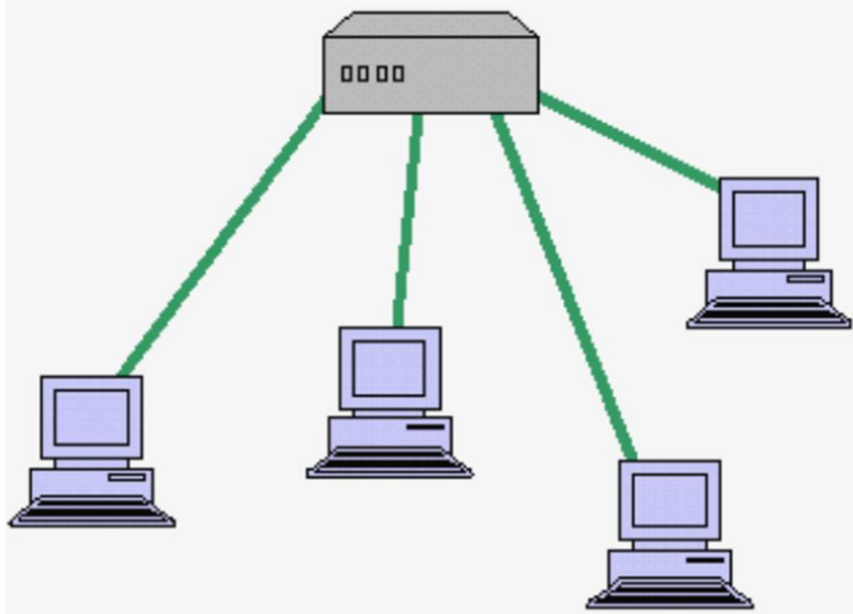
Bus Topology



Ring Topology



Star Topology



Mesh Topology

